

MATHEMATICS

Why the Number e Shows Up Everywhere

Compound interest, population growth, radioactive decay, and probability all lead to the same constant: 2.71828... Here's the single idea that ties them together.

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The number $e \approx 2.71828$ feels arbitrary at first, like π 's less famous cousin. But e isn't a quirk of geometry — it's the natural language of *anything that grows in proportion to its current size*. Once you see that, it stops being mysterious.

■ Born from greed (compound interest)

Suppose you invest \$1 at 100% annual interest. Compounded once a year, you end with \$2. But compound it more often and something interesting happens:

- Yearly: $(1 + 1/1)^1 = 2.00$
- Monthly: $(1 + 1/12)^{12} \approx 2.613$
- Daily: $(1 + 1/365)^{365} \approx 2.7146$
- Continuously: $\lim_{n \rightarrow \infty} (1 + 1/n)^n = e$

No matter how finely you slice the compounding, the total never runs away to infinity. It converges to e . That limit is the definition of e .

■ The property that makes e special

Here's the magic. The function $f(x) = e^x$ is the unique function (up to scaling) that is **its own derivative**:

$$\frac{d}{dx} e^x = e^x$$

In plain words: its rate of growth at every point equals its current value. That's exactly what "growing in proportion to your current size" means — which is why e appears anywhere that description fits.

■ The same pattern, everywhere

- **Population growth:** more individuals $\rightarrow$ more offspring $\rightarrow$ faster growth. Proportional to current size.

- **Radioactive decay:** more atoms $\rightarrow$ more decays per second. Proportional (negatively) to current size, giving $e^{-\lambda t}$.
- **Cooling coffee:** the bigger the temperature gap, the faster it shrinks.
- **Charging a capacitor, drug metabolism, viral spread (early on)** — same differential equation, same e .

They look like different sciences, but mathematically they're one phenomenon wearing different costumes.

■ A surprise from probability

e also sneaks into combinatorics. If n people each toss their hat into a pile and grab one at random, the probability that *nobody* gets their own hat approaches $1/e \approx 0.368$ as n grows — regardless of how many people there are. The same constant, arriving from a completely different direction.

■ Euler's identity: the encore

Tie e to imaginary numbers and you get arguably the most beautiful equation in mathematics:

$$e^{i\pi} + 1 = 0$$

Five fundamental constants — e , i , π , 1 , and 0 — in one line. It emerges naturally once you extend e^x to the complex plane.

■ The takeaway

e isn't a number someone invented; it's a number the universe keeps insisting on. Any time change is proportional to the current amount, e is already there waiting. That's why it's *natural* — the natural logarithm, natural growth, natural decay all point back to this one constant.

Know another surprising place e turns up? Add it in the comments.